

Thematic Analysis:

Thematic analysis is a **Qualitative Data Analysis Technique**. It is frequently used to describe a group of texts, such as an interview or transcripts. The researcher carefully studies the data to uncover common themes - subjects, concepts, and meaning patterns that appear frequently.

Thematic analysis can be done in various ways, but the most popular method involves six steps:

- Familiarization
- Coding
- Generating themes
- Reviewing themes
- Defining and labeling themes
- Writing up

Following this procedure might also assist us in avoiding confirmation bias when developing the analysis.

Thematic Analysis is a good approach when we have to find people's opinions, knowledge, and experience. For the thematic analysis example, I am considering the data set of COMSAT Students that I collected as surveys in google form regarding factors that affect student performance.

Step 1: Familiarization:

The 1st step of thematic analysis is Familiarization with the collected data set. I have a deep understanding of the data set which is collected from the Computer Science Students. Here is an overview of the data. For my FYP I collected data from COMSAT University Students with the help of google forms. So, I will get to know the opinion of students regarding factors that they think can affect their study performance. Survey questions consist of

- Student Personal Details
- Academic Details
- Social Details

Step 2: Coding:

The data must then be coded. Coding is the process of marking areas of our text - usually phrases or sentences - and creating shorthand labels or "codes" to explain their content. Here are the codes which I gathered from large data set of students. According to the provided data set these

are the factors that affects students' performance. So, I keep the coded names which I gathered from large paragraph raw data/statements. Along with it the time these words repeated also mentioned.

Attributes	Frequencies
Mental Health Condition	41
Physical Health Condition	14
Study Hours	42
Traveling Time	17
Interest In The Domain	67
Social Activity Time	30
Teacher Behaviour/Study Method	22
Depression Issues	2
Sleeping Hours	1
Environment	6
Diet	1
Teacher Behaviour/Study Method	23
Freelancing	1
Family Issues	3
Mobile Usage	3
Motivation	4
seriousness level	8
Irrelevant Subjects	1
time management	3
family support	1
Family/Study Background	4
Financial Issues	5
personal Reasons	2
Doing Daily Work	5

Step 3: Generating themes:

Next, I look over the codes I have created, identify patterns among them, and start coming up with themes.

Themes are generally broader than codes. Most of the time, you'll combine several codes into a single theme. In this example, I might start combining codes into themes like this:

Codes	Themes
● Family Issues ● Family Background ● Family Support	Family History
● Mental Health ● Physical Health ● Diet ● Depression ● Stress	Health Issues
● Social Activities ● Study Hours ● Freelancing ● Time Management ● Interest In Domain ● Study Method ● Mobile Usage	Personal Reasons

Here some of the codes categorized in themes. The name of the themes is making sense with the text codes.

Step 4: Reviewing themes:

Now I must make sure that my themes are useful and accurate representations of the data. Here, I return to the data set and compare themes against it. Am I missing anything? Are these themes really present in the data? What can I change to make our themes work better?

If I encounter problems with themes, I might split them up, combine them, discard them, or create new ones: whatever makes them more useful and accurate.

In this example after reviewing, I think personal reasons name can be changed into **Personal Grounds** as I think it's more relatable to the codes.

Step 5: Defining and naming themes:

Defining themes involves explaining precisely what we mean by each theme and determining how it contributes to our understanding of the data.

The process of naming themes includes coming up with a concise and easily accessible name for each theme.

Step 6: Writing up:

Finally, I write up my data analysis. Like in my example I come to **1st result as**

Most students think **Interest in The Domain** is the main reason that positively and negatively can affect the performance of the students. Like if they don't like their degree and they are choosing it because of the family pressure they will not perform good in it and just try to get rid of it. But on the other hand, if students are choosing Software Engineering/Computer Science because of interest in this field their performance will be better from other students, and they are always willing to learn new things and technology time to time.

My 2nd conclusion is students also think less **Study Hours** also affect performance of students. If someone not giving proper time to study, then automatically their performance going to affect. On the other hand, students who give more time to study and are hardworking they can get good grades because of their effort on study.